XYZ LAYOUT ENGINE · AN 18-MINUTE TALK

STOP RE-ROLLING THE WHOLE IMAGE

Deterministic infographics with XYZ Layout Engine

Generative image models can make a great-looking infographic. Fixing one label or moving one callout means regenerating the whole thing and hoping nothing else changes.

XYZ Layout Engine splits the work: AI handles art direction, copy and isolated cutout illustrations; a deterministic renderer owns exact geometry, typography and export.

Every slide in this deck is a JSON fixture rendered by the engine's own renderer in both backends, Satori + resvg and headless Chromium. No slide is an image-model output.

AI-owned Engine-owned

THE RE-ROLL TAX ON ONE LABEL



Change only the headline.

WITH AN IMAGE MODEL

Regenerate the whole image and hope. The leaves, icons, spacing and every other line of copy can drift, and the result is never the same twice.

WITH XYZ LAYOUT ENGINE

Edit one field in fixture.json and rerender. Every other element stays where it was, byte for byte, and the same input gives the same output.

Shown: the nutrition infographic (nutrition@4.0.0), rendered by the engine at 1000 × 1000. Its seven illustrations are AI-generated cutouts (gpt-image-2.5-flare); the layout and type are not.

AI-owned Engine-owned

SPLIT THE WORK: TASTE VS GEOMETRY

AI OWNS

Taste and art direction

Copy

Isolated cutouts: true-alpha PNG

or inline vector SVG

From a DESIGN.md brief or a reference image. Done once per design.

THE HAND-OFF

tools/recipes/<slug>.mjs

fixture.json

cutouts

a versioned recipe, a schema-validated fixture and pinned artwork: files on disk

THE ENGINE OWNS

Exact geometry

Typography from bundled fonts

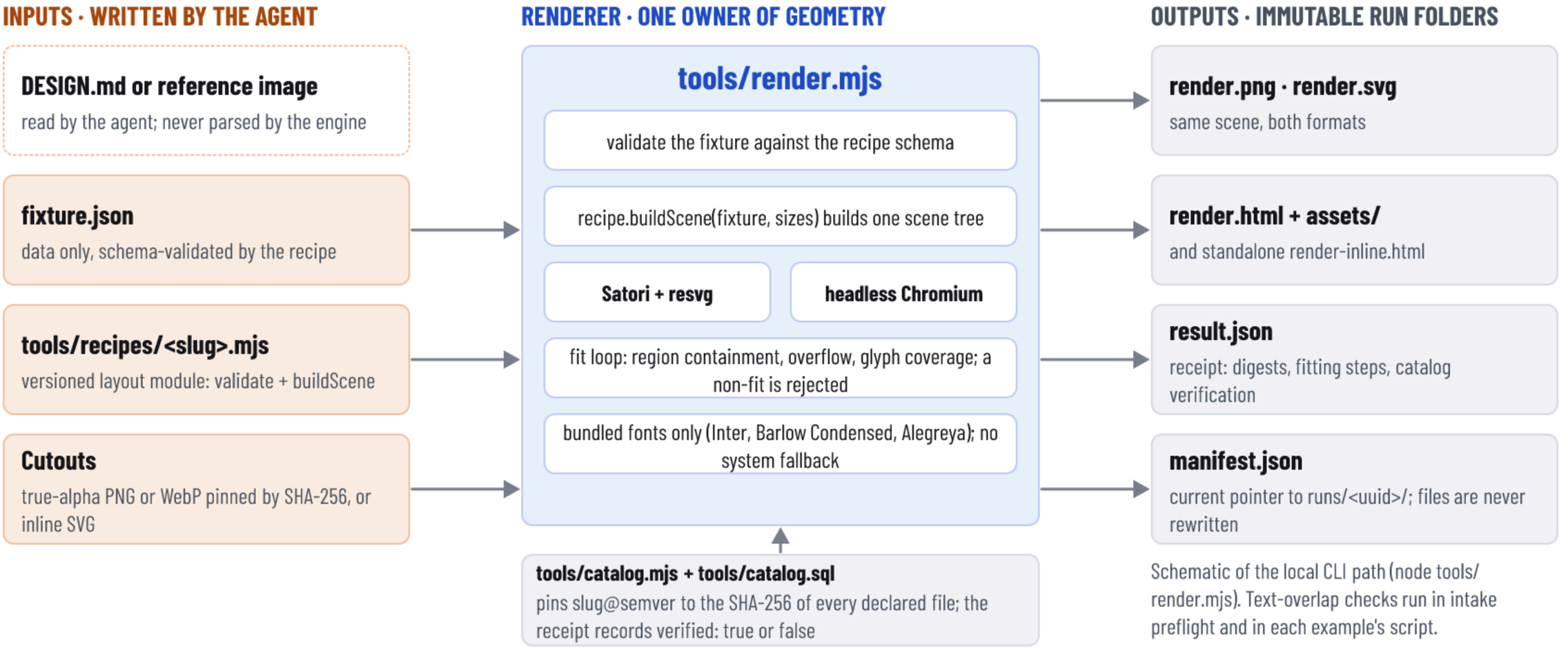
Collision and clipping checks

Export: PNG, SVG, HTML + manifest.json

Offline and deterministic, with no AI at runtime. Done on every render.

AI-owned Engine-owned

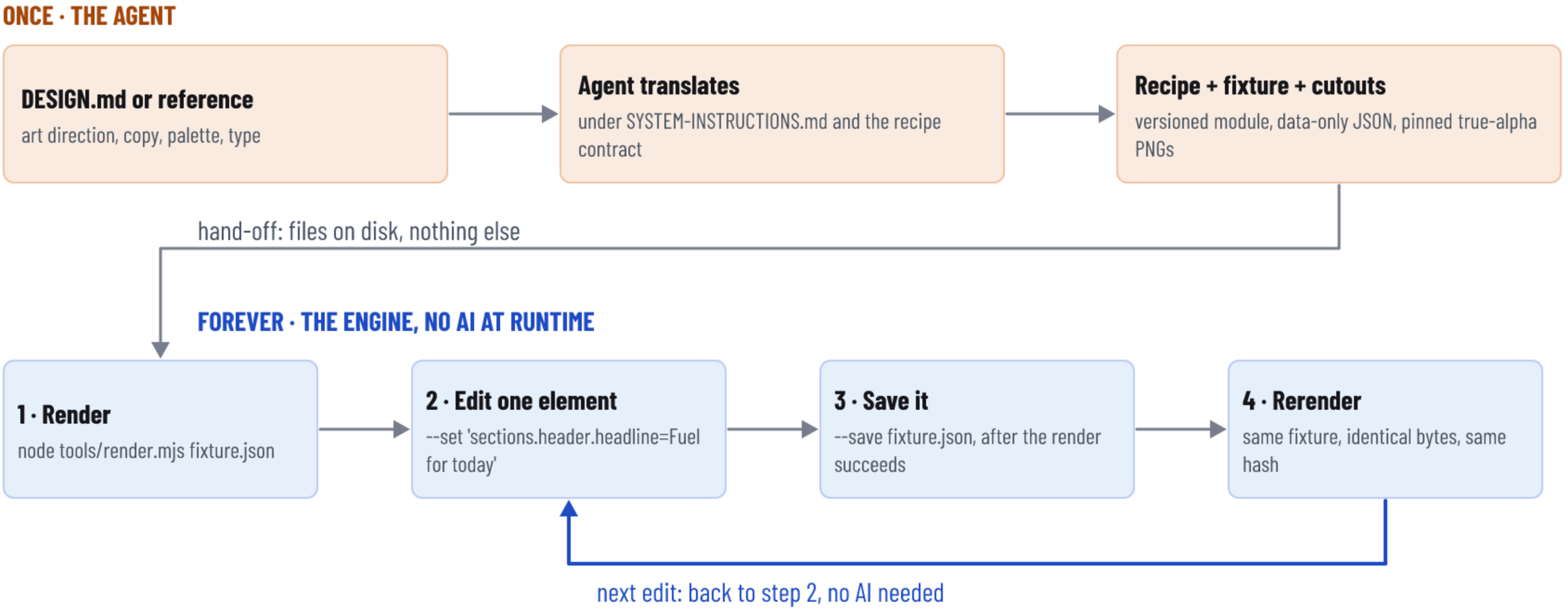
SYSTEM: INPUTS, RENDERER, OUTPUTS



Schematic of the local CLI path (node tools/render.mjs). Text-overlap checks run in intake preflight and in each example's script.

AI-owned Engine-owned

AUTHOR ONCE, EDIT FOREVER



LIVE DEMO: EDIT ONE ELEMENT

1 · PREPARE A SCRATCH COPY INSIDE THE REPO

a fixture under /tmp is refused as a symlink escape

```
mkdir -p .relay-scratch/talk-demo
cp tools/spike/fixture.json .relay-scratch/talk-demo/n.json
```

2 · RENDER THE BEFORE

PNG and SVG into an immutable run folder

```
node tools/render.mjs .relay-scratch/talk-demo/n.json --format png,svg --out .relay-scratch/talk-demo/before
```

3 · EDIT TWO FIELDS, SAVE, RENDER THE AFTER

--set edits are schema-validated; --save writes the fixture only after the render and export succeed

```
node tools/render.mjs .relay-scratch/talk-demo/n.json \
  --set 'sections.header.headline=Fuel for today' --set 'theme.palette.primary=#335577' \
  --save .relay-scratch/talk-demo/n.json --format png,svg --out .relay-scratch/talk-demo/after
```

4 · RERENDER AND COMPARE HASHES

the saved fixture renders to the same SVG bytes

```
node tools/render.mjs .relay-scratch/talk-demo/n.json --format svg --out .relay-scratch/talk-demo/again
shasum .relay-scratch/talk-demo/{after,again}/runs/*/render.svg
```

What to watch: only the headline text and the primary color change. Every other element stays byte-identical, and the after and again SVGs hash the same.

Node 22.2 or newer (verified on v26.0.0; Node 18 fails on node:zlib crc32). Dependencies: pnpm install --frozen-lockfile.

4:00 – 8:00 · THE RESULT (PRE-RENDERED FALLBACK)

TWO FIELDS CHANGED. NOTHING ELSE.



BEFORE · FUEL WELL. LIVE WELL. · #2f5d27



AFTER · FUEL FOR TODAY · #335577

Real renders from the demo commands (nutrition@4.0.0, Satori + resvg, 1000 × 1000). The illustrations inside them are AI-generated cutouts.

CHANGED

sections.header.headline and theme.palette.primary

UNCHANGED

layout, artwork, type scale and every other field

SAME HASH

after and again render.svg, shasum: ceefd615 f5ca0c4e 2a7c80c1 ce947d7d 3a9dc6c2

TIME

under a second per command, Node startup included (0.39 to 0.61 s measured on this Mac, Node v26)

RECIPES: VERSIONED, PINNED, HASHED

A recipe is a versioned layout module.

tools/recipes/nutrition.mjs exports name = 'nutrition' and version = '4.0.0'. It validates the fixture and builds the scene.

Published versions are pinned by hash.

tools/catalog.sql records slug@semver and the SHA-256 of every declared file. node tools/catalog.mjs verify fails on any drift.

A change ships as a new version.

Published rows are immutable, so past renders stay reproducible. Every receipt records the recipe version and verified: true or false.

The fixture is data, never code.

Schema-validated: unknown fields, code and remote assets are refused. A layout that does not fit is rejected, not shipped.

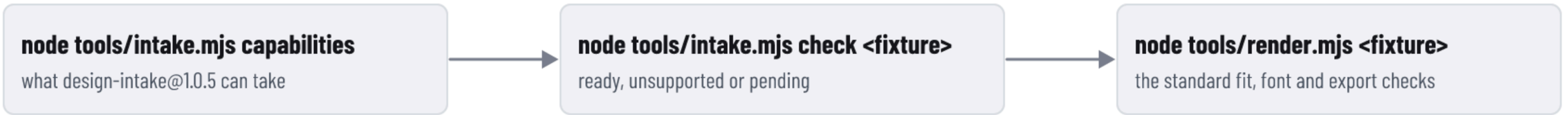
ACTIVE RECIPES IN tools/catalog.sql

slug@version	content SHA-256, first 12
design-intake@1.0.5	1d4399a6bddc
nutrition@4.0.0	c6499955ac44
recursive-process@6.0.0	6995256de110
solar-system@1.0.1	0ed16cdc7b51

From node tools/catalog.mjs show <slug> --json on 2026-10-29. Older versions stay in the ledger as history.

DESIGN.md IS TRANSLATED, NOT PARSED

Google Labs' alpha DESIGN.md format: ordered design rationale plus optional YAML tokens. An agent translates it into settings, artwork prompts and a data-only fixture. The engine never reads the Markdown; it pins the source hash and checks that every requirement quotes it.



THE LINE-HEIGHT STORY

A 1.1 headline line height put the glyphs above their box. The Chromium render refused the clipped text. The fix was in the data-only recipe (the fixture's display line height), 1.1 to 1.3. No fit guard was loosened.

BLOCKED, NOT SIMPLIFIED

A brief the engine cannot support (a second font family, gradients, arbitrary CSS) comes back as a blocked intake with the unsupported fields named, never as a silently simplified render.

Worked example: examples/2026-10-12-design-intake/ (DESIGN.md, fixture.json, both backend renders)

15:00 – 18:00 · CLOSE

THREE WAYS IN, THREE TAKEAWAYS

THREE WAYS TO DRIVE IT

1 · Cloud agent + generated cutouts

authors the recipe and the DESIGN.md contract, generates true-alpha PNGs (SOP.md) and pins them in the example bundle

2 · 100% local LLM, offline

reads SYSTEM-INSTRUCTIONS.md and DESIGN.md, edits fixture.json, self-corrects from the renderer's overflow and overlap findings, draws inline SVG

3 · CI job or backend, no AI at runtime

renders, validates and edits with --set / --save: no network, no paid calls, the same bytes every time

TAKEAWAYS

AI designs once.

Every later edit touches one element.

Recipes and DESIGN.md make the design durable input, not a one-shot prompt.

Deck source: examples/2026-10-29-talk-deck/ · every slide here is a fixture rendered by the engine